

Topic: Introduction to TOC

Notes

Coal India Management Trainee (CS) Recruitment 2026

Contains 2 PDF lessons

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Module

Paper-2 | Theory of Computation

Topic

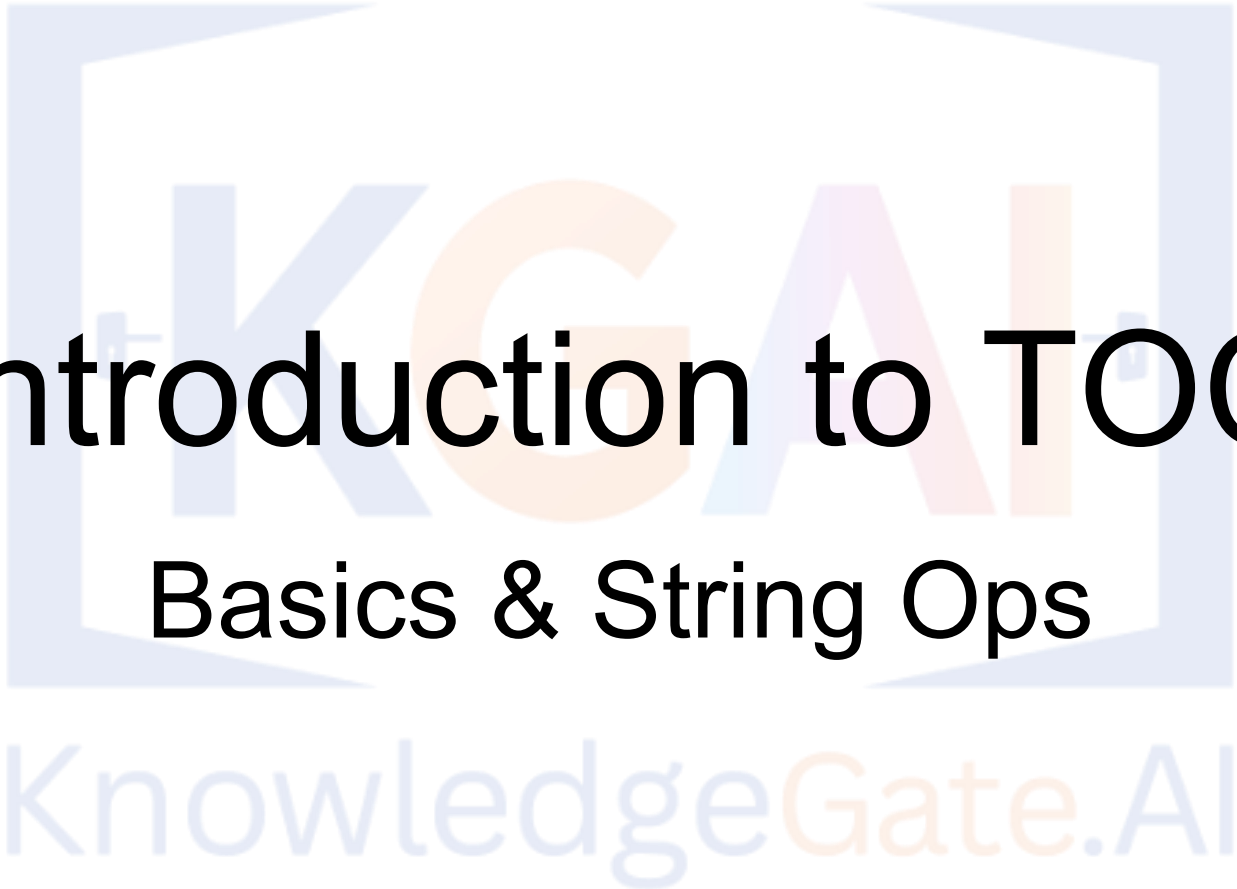
Introduction to TOC

Subtopic

Basics & String Ops

Lesson

Basics & String Ops

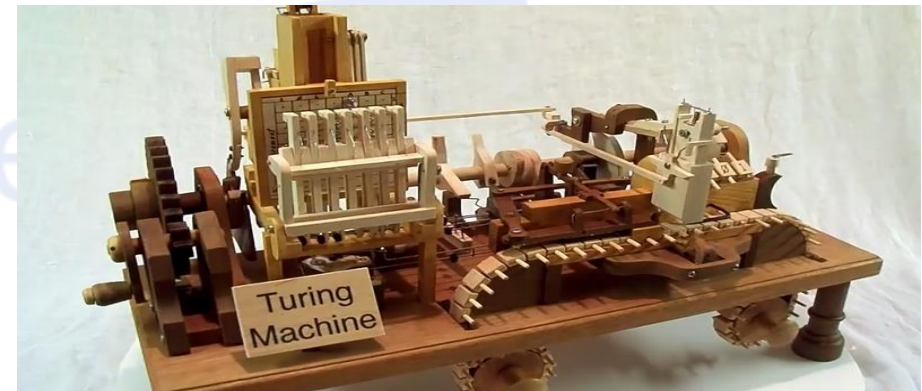
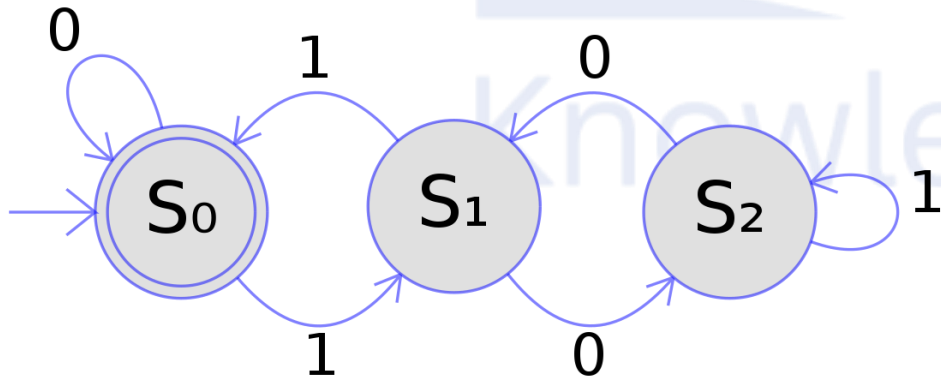
The background features a large, faint watermark of the KnowledgeGate.AI logo. It consists of a blue square frame containing the letters 'KGA' in a stylized font. The 'K' is blue, the 'G' is orange, and the 'A' is pink. Below the frame, the text 'KnowledgeGate.AI' is written in a light blue font, with 'Gate' in orange.

Introduction to TOC

Basics & String Ops

INTRODUCTION TO THEORY OF COMPUTATIONS

- **Theory of Computation (TOC)** is a branch of theoretical computer science that addresses:
- **What can be computed, How efficiently can it be computed, What limitations exist in computation?**
- It involves the study of abstract machines, also known as **automata**, and mathematical models that define what problems can be solved using these machines.
- TOC focuses on different computational models, their capabilities, and the complexity of solving problems. The main areas it deals with include:
 - **Automata Theory**: Studies different types of automata (e.g., finite automata, pushdown automata) and their abilities to solve problems.
 - **Formal Languages**: Explores the classification of languages and grammars that machines can recognize.
 - **Computability Theory**: Investigates which problems can or cannot be solved by any computational model.
 - **Complexity Theory**: Considers the resources (time, space) needed to solve problems and categorizes them based on difficulty (e.g., P, NP).



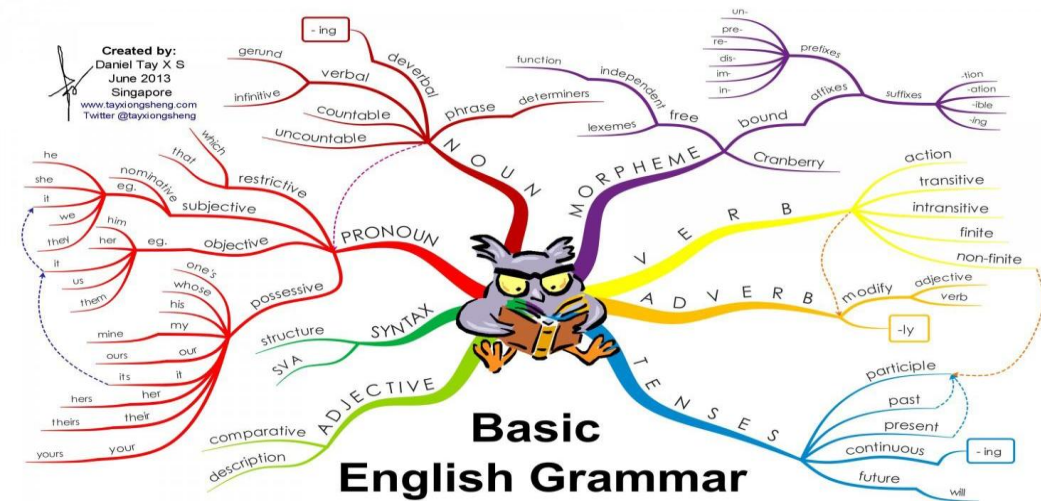
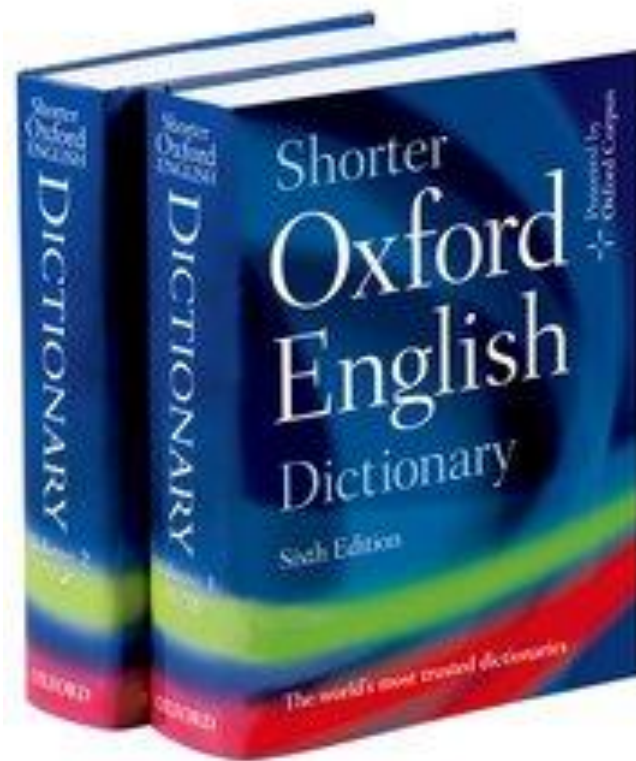
PROBLEM

- **Machine Communication:**
 - Machines play a crucial role in human development, and we need a language to communicate with them. However, this language need not be as complex as natural languages.
- **Formal vs. Informal Languages:**
 - Machines use **formal languages**, which are structured and concise, unlike **informal languages** used in daily human interaction.
- **Language Definition:**
 - A language is "a system suitable for expressing ideas, facts, or concepts" using a set of symbols.



METHODS TO DEFINE LANGUAGE

- In natural language, we can list a finite and predefined set of words in a dictionary, but the sentences formed using these words are infinite. To determine which sentences are valid or invalid, we use **grammar** or rules as a mechanism.



MATHEMATICAL DEFINATION OF LANGUAGE

- **SYMBOL**- Symbols are the basic building blocks, which can be any character/token. (cow, sheep, 😊, white flag, 🚫, 🌟, 🏥 etc.) (in English we called them as letters).
- **ALPHABET**- An alphabet is a finite non empty set of symbols, (every language has its own alphabet). here in toc, we use symbol Σ for depicting alphabet. e.g. $\Sigma = \{0,1\}$. for English $\Sigma = \{a, b, c, \dots, z\}$ (in English also alphabet is a set of letters, thought in general we called them as alphabet).
- **STRING** - It is a finite sequence of symbols (which are the member of set alphabet). E.g. $\Sigma = \{a, b\}$ String- aabb, aa, b, so on. (in English we called them as words).
- **LANGUAGE** - A language is defined as a set of strings. In the next level we consider programs as a string and programming constructs/tokens like int, floats as letters/symbols.

- Similarly, in our system we have finite number of symbols/letters but using those letters we can generate infinite strings/words.
- So, we may have languages that have infinite number of words, so it is not possible for us to list them, we have to use some framework, which can somehow represent the same language. There are mainly two methods to represent a language
 - by a grammar that generates a language [RG generate RL]
 - by a machine that accepts a language [FA accept RL language]

Machine

Grammar

SOME BASIC OPERATIONS ON STRINGS

- So, before we proceed further let's do a little home work on string, which will help us throughout the subject.
- **Length of a string** - It is defined as number symbol in the string. Denoted like $|W|$, e.g. length of string $|00110| = 5$.
- $|aaba| =$
- $|010| =$

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Concatenation of string- Let x and y be two strings, then concatenation is defined as the string formed by making a copy of string x followed by a copy of string y . (NOTE- It's not commutative)

E.g. $w = ab$, $x = ba$

$wx =$

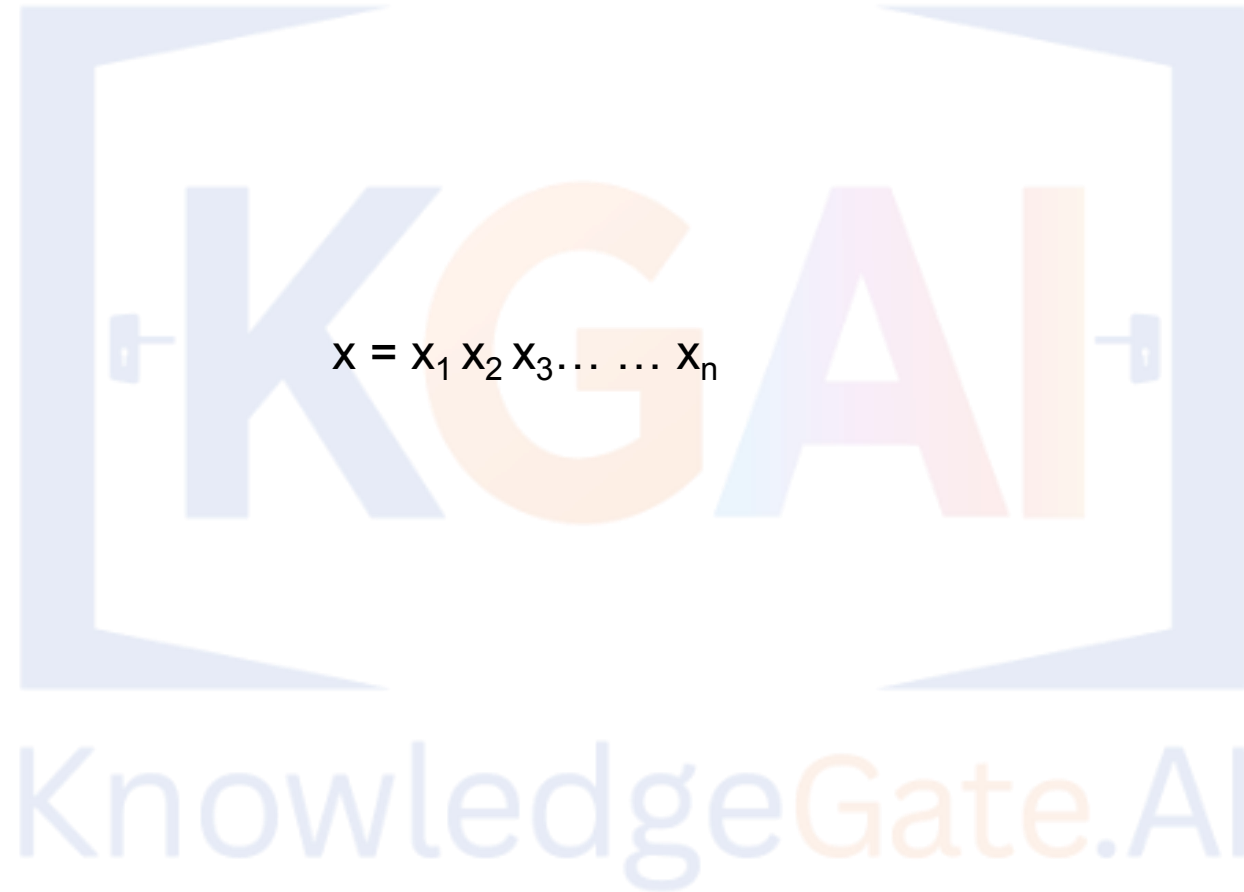
$xw =$

$w = w_1 w_2 w_3 \dots \dots w_m$

$wx =$

$|wx| = |w| + |x|$

$x = x_1 x_2 x_3 \dots \dots x_n$



Reverse of a string – If there is a string w then reverse of a string a denoted by w^r it is just the same string but written in reverse order.

$$W = w_1 w_2 w_3 \dots \dots w_n$$

$$w^r =$$

$$|w| \boxed{} |w^r|$$



Empty/Null String- The string with zero occurrence of symbols. It is denoted by ϵ , $|\epsilon|=0$. If there is a string w , then w^n stands for the strings obtained by repeating w , n times.

$$w^3 = www$$

$$w^2 = ww$$

$$w^1 = w$$

$$w^0 =$$

$$w\epsilon = \epsilon w =$$



Substring- Any string of consecutive symbols in some string 'w' can be collectively said as a substring. E.g. w= abab its substrings can be ab, a, ba...etc.

S u b s t r i n g

- 1. utg
- 2. sbr
- 3. rts
- 4. str
- 5. sub
- 6. ϵ



Q Consider a sting 'GATE' find the total number of substring possible?

	Substring of length 0	Substring of length 1	Substring of length 2	Substring of length 3	Substring of length 4
Number of sting possible	1	4	3	2	1
SUBSTRINGS	€	G, A, T, E	GA, AT, TE	GAT, ATE	GATE

Q Consider a sting 'GGGE' find the total number of substring possible?

	Substring of length 0	Substring of length 1	Substring of length 2	Substring of length 3	Substring of length 4
Number of sting possible					
SUBSTRINGS					

If a string has 'n' distinct symbols then total number of different sub string will be $[n(n+1)/2] + 1$. If w is any string than empty string € and the string w itself is called a trivial substring and the remaining of the other are the non-trivial sub string $[[n(n+1)/2]-1]$

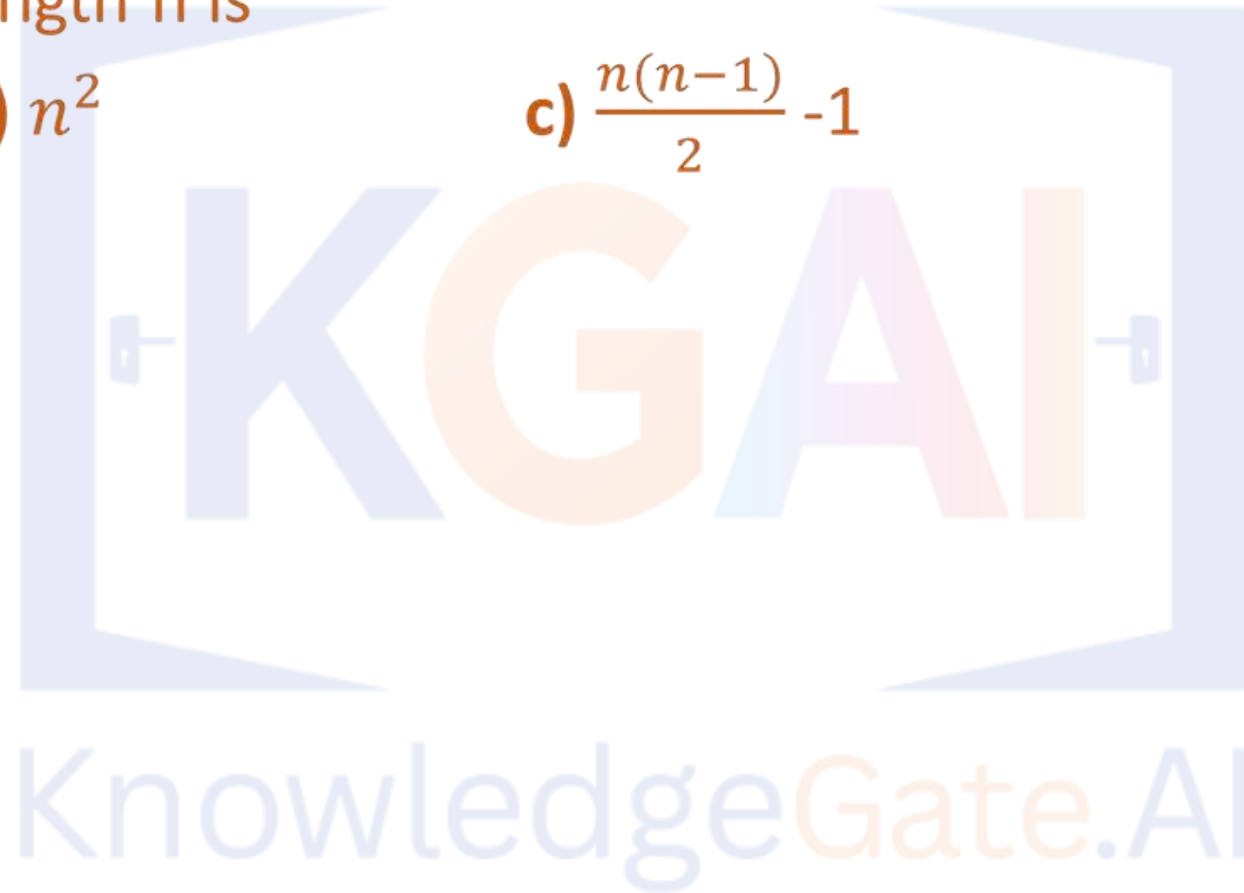
Q The number of substrings (of all lengths inclusive) that can be formed from a character string of length n is

a) $n!$

b) n^2

c) $\frac{n(n-1)}{2} - 1$

d) $\frac{n(n+1)}{2} + 1$

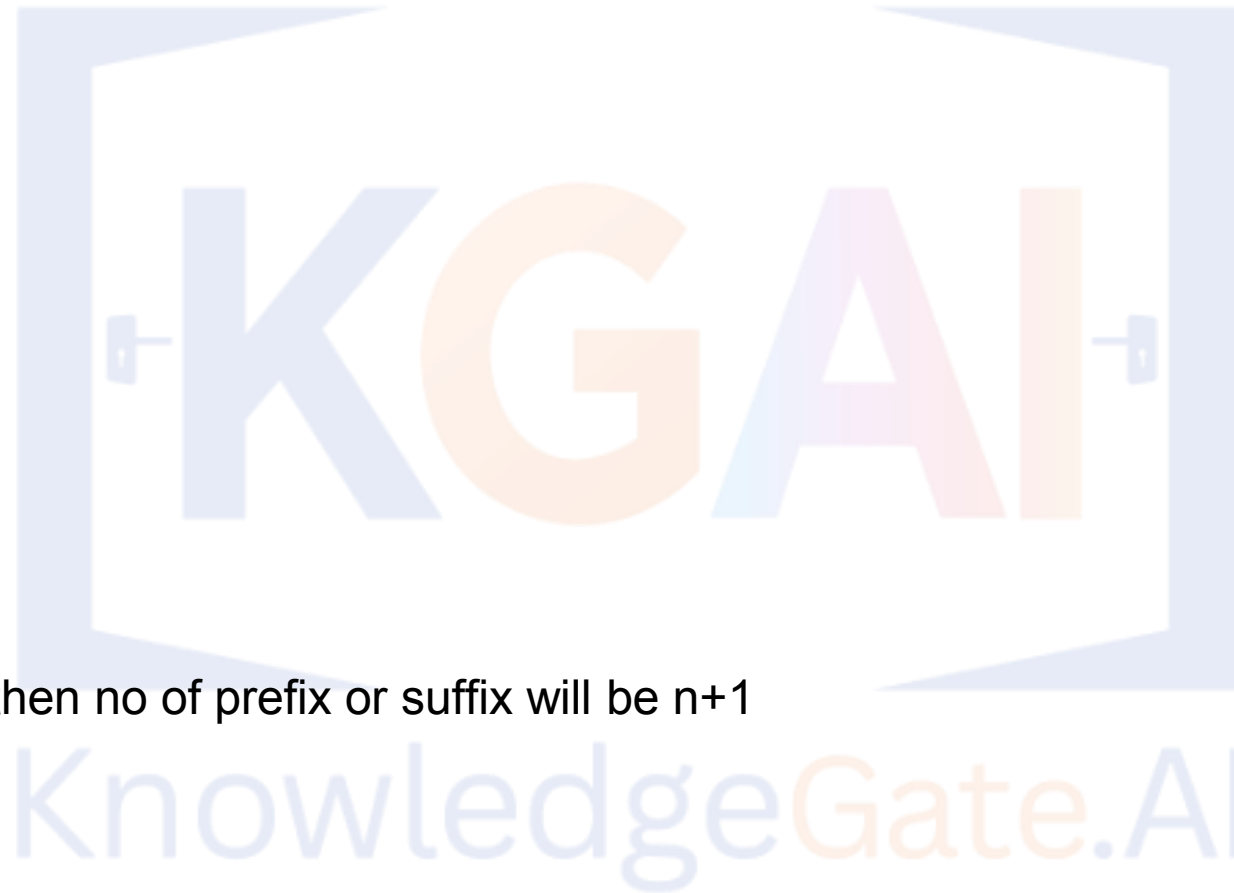


Q Consider a sting 'GATE' find the total number of prefix and suffix possible?

Prefix(GATE) =

Suffix(GATE) =

If there is a string of length n then no of prefix or suffix will be $n+1$



Q In a string of length n , how many proper prefixes can be generated

- a)** 2^n **b)** n **c)** $n(n + 1)/2$ **d)** $n - 1$



Q if $\Sigma = \{a, b\}$ then, find the following?

$$\Sigma^0 =$$

$$\Sigma^1 =$$

$$\Sigma^2 =$$

$$\Sigma^3 =$$

- Σ^K is the set of all the strings from the alphabet Σ of length exactly K.
- $\Sigma^K = \{W \mid |W| = K\}$ (using the symbols from the alphabet Σ)

Kleene closure- If Σ is a set of symbols, then we use Σ^* to denote the set of strings obtained by concatenating zero or more symbols from Σ of any length, in general any string of any length which can have only symbols specified in Σ .

- $\Sigma^* = \bigcup_{i=0}^{\infty} \{w \mid |w| = i\}$ (using the symbols from the alphabet Σ)

Positive closure – If Σ is a set of symbols, then we use Σ^+ to denote the set of strings obtained by concatenating one or more symbols from Σ of any length, in general any string of any length which can have only symbols specified in Σ (except ϵ).

- $\Sigma^+ = \bigcup_{i=1}^{\infty} \{w \mid |w| = i\}$ (using the symbols from the alphabet Σ)

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Q Let x_n denote the number of binary strings of length n that contain no consecutive 0 's, which of the following recurrences does x_n satisfy?

a) $x_n = 2x_{n-1}$

b) $x_n = x_{\left\lfloor \frac{n}{2} \right\rfloor} + 1$

c) $x_n = x_{\left\lfloor \frac{n}{2} \right\rfloor} + n$

d) $x_n = x_{n-1} + x_{n-2}$

Q The value of x_5 is?

a) 5

b) 7

c) 13

d) 16

Module

Paper-2 | Theory of Computation

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Language Ops & Sets

Lesson

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Introduction to TOC

Language Ops & Sets

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LANGUAGES

- Since languages are sets, the union, intersection and difference of two languages are immediately defined.
- The complement of a language is defined with respect of Σ^*
 - $L^c =$
- The reverse of a language is the set of all the strings after reversal
 - $L^r =$
- The concatenation of two languages L_1 and L_2 is the set of all the strings obtained by concatenating any elements of L_1 and L_2 .
 - $L_1 L_2 =$
- Let L^n is defined as L is concatenated with itself n times.
 - $L^n =$
 - $L^* =$

Q Suppose $L_1 = \{10, 1\}$ and $L_2 = \{011, 11\}$. How many elements are there in $L = L_1 L_2$

a) 4

b) 3

c) 2

d) None of these

